

Nisha V Kumar

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EDUCATION

University of Michigan

Master of Science in Engineering in Aerospace Engineering

GPA: 4.0 | Coursework: Combustion, Gas Turbine Propulsion, Hypersonics, Compressible Flow, Numerical Methods

Ann Arbor, Michigan

Expected December 2026

The Ohio State University

Bachelor of Science in Aerospace Engineering

Columbus, Ohio

May 2025

WORK EXPERIENCE

GE Aerospace Research, Thermal Management Systems

Niskayuna, NY

Edison Engineering Intern

May 2026 – Present

- Built a transient two-phase cryogenic hydrogen fuel tank model in Simscape/Simulink to simulate liquid-H₂ delivery, gaseous-H₂ recirculation, and PI-controlled tank pressurization for turbofan fuel system analysis.
- Integrated 6+ thermal subsystems, including recirculation flow paths, vaporization and cooling elements, boil-off behavior, and a 3-layer tank wall heat-transfer network, enabling independent model testing and verification prior to full system integration.
- Simulated tank response across 4 operating phases – ground idle, takeoff, climb, and cruise – using time-varying engine demand profiles to evaluate system operability and transient behavior for next-generation cryogenic hydrogen fuel architectures.

Honeywell Aerospace, Engines & Power Systems

Phoenix, AZ

Engine Performance Intern

May 2024 – August 2024

- Generated $\pm 2\%$ sensitivity datasets for the AGT1500 turboshaft engine, improving thermodynamic cycle-model fidelity and supporting low-power certification of U.S. Army Abrams tank engines.
- Investigated high-power compressor surge by reproducing field power-lever profiles in a test cell, generating compressor-map data that supported root-cause and corrective-action analysis and surge-prevention measures at 1,500 SHP.

Maxar Technologies, Space Division

Palo Alto, CA

Quality Systems Intern

May 2023 – July 2023

- Identified and resolved 20+ nonconformances across satellite high bays and test chambers, supporting ISO 9001/AS9100 compliance for \$1B+ spacecraft programs, including Jupiter 3, Galaxy 37, and WorldView Legion.

RESEARCH & PROJECT EXPERIENCE

University of Michigan, Gas Dynamics Imaging Laboratory

Ann Arbor, MI

Graduate Research Assistant || Advisor: Professor Mirko Gamba

August 2025 – Present

- Developed a two-region, quasi-one-dimensional MATLAB model combining Fanno flow and separated-flow physics to reproduce shock-train and downstream mixing pressure rise behavior across Mach 2 scramjet isolator experiments.
- Eliminated case-by-case shock-train calibration by deriving a scaling relation based on local Mach and Reynolds numbers that predicted iteratively fitted shock-train parameters within a 2.9% mean difference and improved reduced-order analysis.
- Selected for first-author presentation of this research at the 2027 AIAA SciTech Forum in Orlando, FL.

OSU Competitive Rocketry Team: Buckeye Space Launch Initiative

Columbus, OH

NASA Deputy Project Manager & Safety Officer

August 2021 – May 2025

- Directed 30+ students across structures, payload, flight systems, and testing through the full design-build-test-flight lifecycle of a 10.5-ft, 6-in diameter, L-class solid motor rocket and data-transmission payload for NASA Student Launch.
- Led mission safety and launch readiness through FMEA and personnel-hazard analyses, OpenRocket flight simulations, recovery planning, and launch procedures, enabling safe construction, testing, and competition flight at NASA MSFC, Huntsville, AL.

OSU Experimental Senior Capstone: AFRL JetCat P100-RX Turbine Engine

Columbus, OH

- Placed 4th among 19 universities in the 2024–25 Aerospace Propulsion Outreach Program by designing and testing a linearly actuated variable-area nozzle for a JetCat P100-RX engine, integrating load-cell instrumentation, engine controls, LabVIEW data acquisition, and ANSYS flow analysis, with final competition testing at the Air Force Research Laboratory.

OSU Turbine Aerothermodynamics Lab, Aerospace Research Center

Columbus, OH

Undergraduate Research Assistant || Advisor: Dr. Jeffrey P. Bons

May 2022 – July 2022

- Executed high-temperature kiln-rig tests to evaluate effusion-cooling degradation in gas-turbine combustor liners; applied microscopy, 3D imaging, and MATLAB to quantify capture efficiency & deposition height and standardized test protocol.
- Replicated deposition conditions on Hastelloy X effusion plates across multiple particle compositions, demonstrating 37% less hole blockage from synthetic dust than industry mixtures; results published at AIAA SciTech 2023.

SKILLS

Design & Modeling: MATLAB, Simulink, Simscape, SOLIDWORKS, OpenRocket

Testing & Analysis: LabVIEW, ANSYS Fluent, DAQ systems

Fabrication & Characterization: 3D Printing, Spot Welding, Soldering, Laser Cutting, 3D Imaging, Microscopy, Particle-Size Analysis